Ecologic Studies and Confounding

APHS 20203 · Epidemiology & Biostatistics

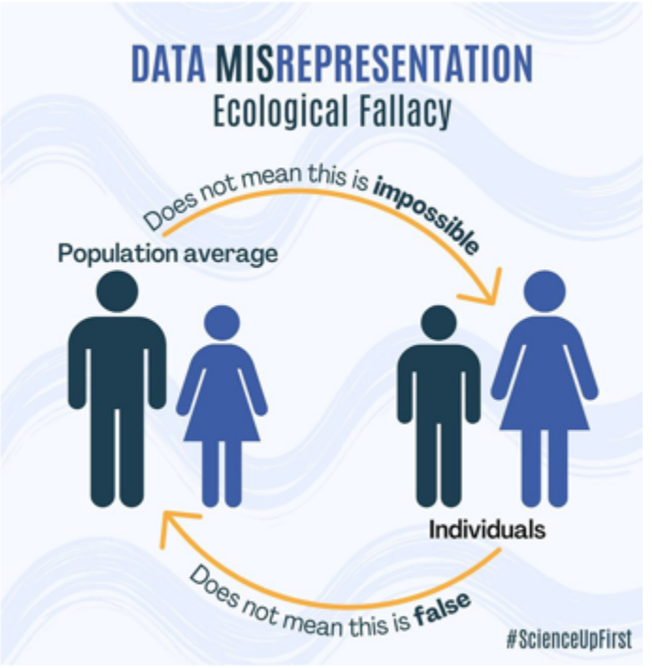
October 14, 2026

Ecologic Studies

- **Ecologic Studies:** *Group-level analysis*
- **Unit of analysis:** groups (not individuals)
- **Exposure & outcome:** measured at the **population level**
- **Common use:** explore population patterns; generate hypotheses
- **Key caution:** group-level associations ≠ individual-level conclusions
- **Take-home message:** Ecologic studies are useful for **population questions**, but **cannot establish individual-level causality**.

Ecologic Fallacy

- **Definition:** Incorrectly applying **group-level** findings to **individuals**
  - Occurs when associations observed in populations are assumed to hold at the individual level
- **Example**
  - Observation: Cities with higher income have higher CHD rates
  - Incorrect inference: Wealthier individuals have higher CHD risk
  - Reality: Within cities, CHD risk may be higher among lower-income individuals
- **Take-home message:** Group-level data cannot be used to draw **individual-level conclusions**.



https://www.scienceupfirst.com/project/data-misrepresentation-ecological-fallacy/

Ecologic Study: Strengths & Limitations

Strengths

- Uses existing population-level data
- Efficient: fast and low cost
- Useful when individual-level data are unavailable

Limitations

- **Ecologic fallacy**
- Crude exposure measurement
- Limited control of confounding

Dietary Fat and Breast Cancer Mortality

- **Study**
  - Sasaki et al. (1993): dietary fat intake & breast cancer mortality
  - Unit of analysis: **countries (30 countries total)**
- **Key finding**
  - Higher animal fat intake was associated with higher breast cancer mortality
  - Fish fat intake was associated with lower breast cancer mortality
- **Discussion prompts**
  - What makes this study **feasible and efficient**?
  - Why can these results **not** be interpreted at the individual level?
  - What **other population-level factors** could explain the association?

Sasaki S, Horacsek M, Kesteloot H. An ecological study of the relationship between dietary fat intake and breast cancer mortality. Prev Med. 1993 Mar;22(2):187-202. doi: 10.1006/pmed.1993.1016. PMID: 8483858.

Low-Fat Dietary Pattern: Randomized Trial

- **Study overview**
  - Large randomized controlled trial of a **low-fat dietary pattern**
  - Postmenopausal women assigned to **intervention** or **usual diet**
  - Long-term follow-up
- **Discussion prompts**
  - Why is an **RCT** an appropriate design for this question?
  - What features of this study strengthen **causal inference**?
  - What challenges might arise in **long-term behavioral interventions**?
- **Key takeaway:** RCTs strengthen causal conclusions, but feasibility, adherence, and follow-up matter.

Prentice RL, Caan B, Chlebowski RT, et al. Low-Fat Dietary Pattern and Risk of Invasive Breast Cancer: The Women's Health Initiative Randomized Controlled Dietary Modification Trial. JAMA. 2006;295(6):629–642. doi:10.1001/jama.295.6.629

Group Discussion: Comparing Study Designs

- **Comparing Two Studies:** Think back to the two studies discussed today:
  - **Ecologic study:** dietary fat intake and breast cancer mortality
  - **Randomized controlled trial:** low-fat dietary intervention
- **Discussion questions**
  - What is the **unit of analysis** in each study?
  - Which study provides **stronger evidence for causality**, and why?
  - What kinds of conclusions are **appropriate** for each design?

Social Desirability Bias

- **Definition:** participants provide responses that are viewed as **socially acceptable** rather than accurate.
- **When it occurs:** sensitive behaviors, such as smoking, alcohol use, drug use, and other self-reported behaviors.
- **Key idea:** When participants modify responses to appear favorable, exposure or outcome data may be systematically distorted.
- **Reducing it:** Ask about behaviors in a **neutral, nonjudgmental** way.

Hawthorne Effect

- **Definition:** participants change behavior because they know they are being observed.
- **When it occurs:** during clinical trials or behavioral studies, when participants are monitored or aware of measurement.
- **Key idea:** When behavior changes due to observation, the measured exposure or outcome may not reflect usual behavior.

Exercise: Bias and Behavior Changes

For each scenario, identify the **bias or behavior change** being illustrated and explain why.

- In a case-control study, parents of sick children recall pesticide exposure during pregnancy more carefully than parents of healthy children.
- Participants report lower alcohol use in face-to-face interviews than expected from sales data.
- In a clinical trial, outcome assessors know which participants received the treatment and rate them more favorably.
- Interviewers ask more detailed exposure questions to lung cancer cases than to controls.
- Participants increase exercise levels during a study because they know their activity is being monitored.

Reducing Bias

- **Reducing Selection Bias**
  - Use appropriate **comparison groups**
  - Apply clear and consistent **inclusion criteria**
  - Minimize **loss to follow-up**
  - Ensure recruitment is not related to both exposure and outcome
- **Reducing Information Bias**
  - Use **standardized definitions** and procedures
  - Apply **blinding** when possible
  - Collect data **consistently across groups**
  - Use validated measurement tools
- **Key idea:** Careful design and consistent measurement strengthen internal validity.

Review: Sample Size and Bias

- **Random error:** Natural variability in data; affects **precision** and is generally reduced by increasing sample size.
- **Bias:** Systematic error arising from study design, selection, or measurement. A larger sample does **not generally eliminate bias**.
- **Existing example:** Measuring physical activity at local gyms may systematically overrepresent more active people, even with a large sample.

For information bias: standardized definitions, consistent measurement procedures, and blinding address the measurement process; a larger sample alone does not.

Review: Confounding

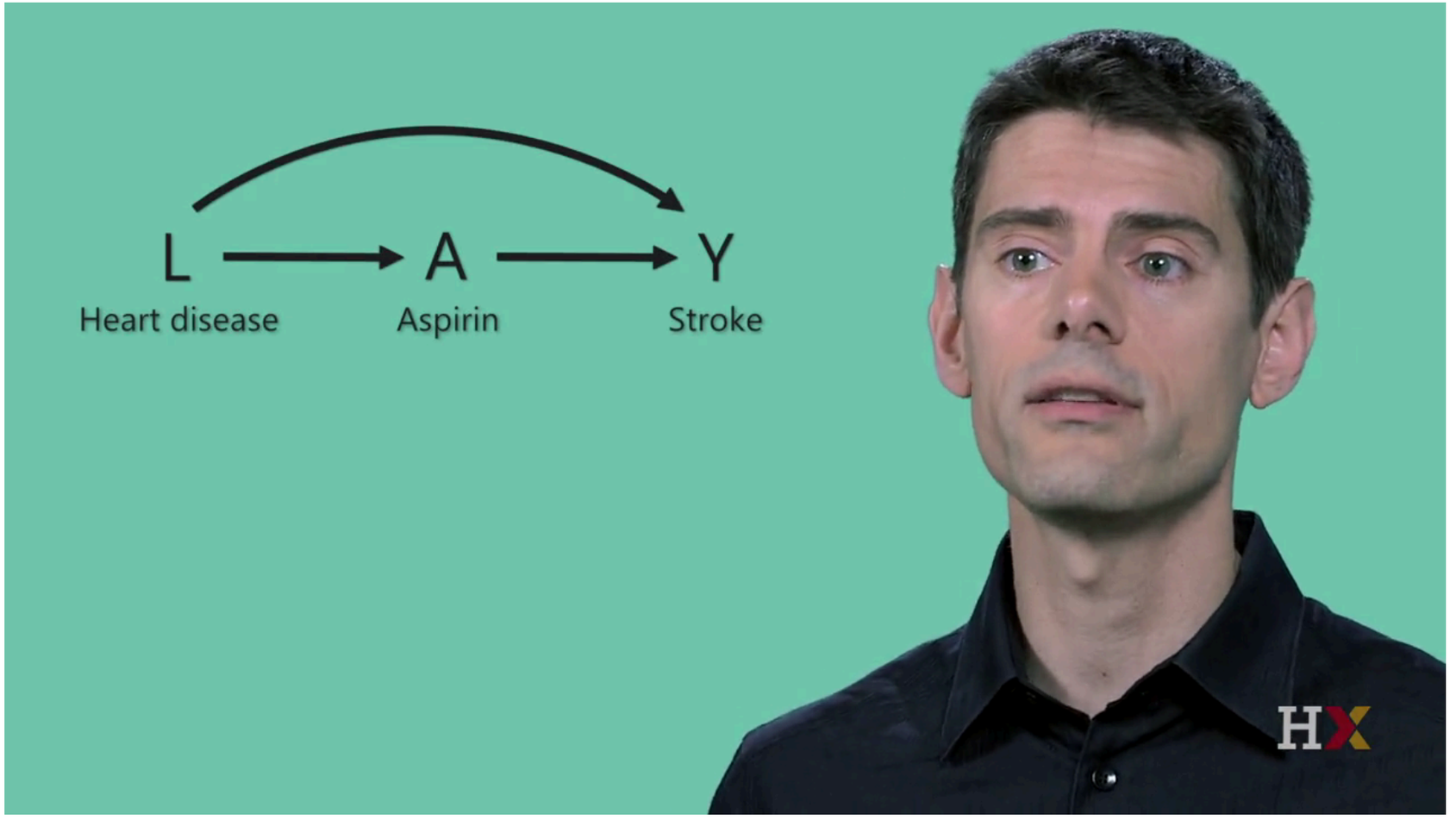
- **Confounding:** The effects of other factors become mixed into our estimate of an exposure's causal effect. Or giving credit to the wrong variable.
- **Example:** Ice cream sales and drownings both increase during hot weather.
  - Hot weather influences both ice cream sales and drowning risk.
- **Question:** Would reducing ice cream sales reduce drownings?
- **Key idea:** The association can be real and useful for prediction without representing a causal effect of ice cream sales.

Conventional Confounder Definition

L is a confounder of the effect of A on Y if L meets these three conditions:

- L is associated with A
- L is associated with Y conditional on A
- L is not on a causal pathway from A to Y

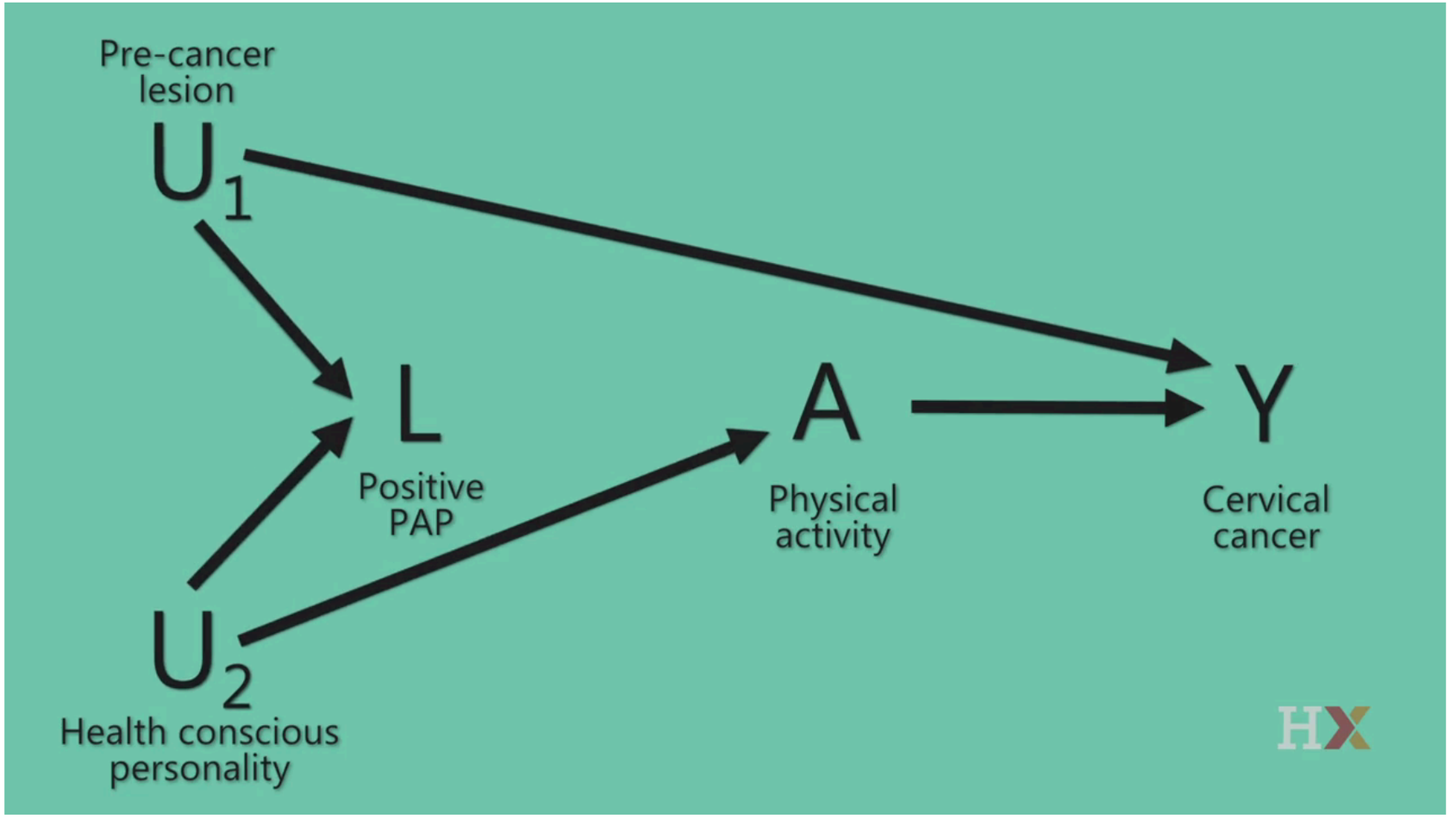
Confounders: Part 1



Confounders: Part 3



Confounders: Part 3 Diagram



Poll: Confounder Criteria

Consider a variable associated with the treatment A, associated with the outcome Y within levels of A, and not on a causal pathway from A to Y. Adjustment for such a variable will always reduce bias.

A. True

B. False

Poll: Causal Structure

Generally, we need to use our expert knowledge of the causal structure of a problem to determine whether confounding is present.

A. True

B. False

Addressing Confounding

- Confounding cannot be ignored when interpreting results.
  - **Randomization** helps balance confounders between groups
  - In observational studies, researchers must **identify and measure potential confounders**
  - Results should be interpreted cautiously if important confounders are not accounted for
- **Key idea:** If confounding is not addressed, an observed association may not reflect a true causal relationship.

Review: Cohort vs. Case-Control

Feature	Cohort Study	Case-Control Study
Start With	Exposure	Outcome
Direction	Exposure → Outcome	Outcome → Exposure
Measure	Relative Risk (RR)	Odds Ratio (OR)
Best For	Rare exposures	Rare diseases
Time & Cost	Longer, more expensive	Faster, less expensive
Main Bias Concern	Loss to follow-up	Selection & recall bias
Can Measure Incidence?	Yes	No

Epidemiology Wrap-Up

- Define and explain the **foundations of epidemiology**
- Measure disease using **incidence, prevalence, and rates**
- Describe patterns by **person, place, and time**
- Evaluate **association and causality**
- Distinguish **cohort and case-control** study designs
- Interpret **RR, ARD, and OR**
- Identify major sources of **bias and confounding**

From Concepts to Your Presentation

- **The Conceptual Progression**
  - Descriptive Epidemiology
    → Generates hypotheses
  - Analytic Epidemiology
    → Tests exposure → outcome relationships
  - Critical Thinking
    → Distinguish **association from causation**

- **For Your Epi Final Presentation**
- You are expected to:
  - Clearly describe a population health issue
  - Include and interpret an **analytic study**
  - Identify the **study design and measure**
  - Discuss **bias, confounding, and limitations**
  - Avoid overstating causality